

SPLUNK SIEM SECURITY MONITORING & INCIDENT INVESTIGATION

A Practical Security Monitoring and Privileged Activity Investigation

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Environment: Kali Linux—VirtualBox—Splunk Enterprise 10.4.3

Project Type: SIEM / SOC Security Monitoring & Incident
Investigation

Assessment Environment: Authorized Local Cybersecurity
Laboratory

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1. EXECUTIVE SUMMARY

This project demonstrates a practical Security Information and Event Management (SIEM) monitoring and investigation workflow using Splunk Enterprise in an authorized local cybersecurity laboratory environment.

Splunk Enterprise 10.4.3 was deployed on a Kali Linux virtual machine running in VirtualBox. A security-log dataset containing sudo and PAM authentication activity was collected from the Linux system journal and ingested into a dedicated Splunk index named "splunk_lab".

The investigation focused on identifying, analysing, and understanding privileged activity within the laboratory environment. Splunk Search Processing Language (SPL) was used to extract user and account information, identify privileged commands, analyse activity over time, investigate periods of increased activity, and examine system and security-tool administration events.

A total of 215 security events were successfully ingested into the Splunk laboratory index. Analysis identified 67 root-targeted privilege events involving 36 distinct commands. The observed activity included system service management, package management, system reboots, system-log inspection, Nessus security-tool administration, and user-group modification.

A Splunk detection alert titled "Root Privilege Activity Detection" was also created and enabled to identify root-level privilege activity. The alert was configured as a real-time detection with a trigger condition of more than zero matching results within one minute. Trigger history demonstrated that the detection was being activated by matching activity.

The investigation found that the observed privileged activity was predominantly consistent with authorized laboratory administration, system maintenance, and cybersecurity-tool configuration. Based on the available dataset and investigation performed, no confirmed malicious activity was established.

This project demonstrates practical junior SOC analyst capabilities in SIEM deployment, security-log ingestion, SPL querying, privileged-activity monitoring, timeline analysis, detection engineering, alert investigation, evidence collection, and security reporting.

2. Assessment Environment

The assessment was conducted in an authorized local cybersecurity laboratory designed to simulate a security monitoring and investigation workflow.

2.1 Host and Virtualization Environment

- Host Operating System: Windows 10 Pro
- Virtualization Platform: Oracle VirtualBox
- Security Testing Operating System: Kali Linux
- Kali Linux Environment: Virtual machine
- Virtual Machine Memory: 4 GB RAM
- Splunk Web Interface: Port 8000
- Splunk Installation Path: "/opt/splunk"

2.2 SIEM Platform

- Platform: Splunk Enterprise
- Version: 10.4.3
- Build: 4174a2deda5d
- Operating Account: "splunk"
- Primary Laboratory Index: "splunk_lab"

2.3 Log Dataset

The investigation used a static security-log dataset generated from the Kali Linux system journal.

The dataset was collected using "journalctl" and filtered to capture sudo-related authentication and privilege activity. The resulting log file was stored at:

`"/var/log/splunk-lab/splunk_auth.log"`

The dataset contained 215 events and included information such as initiating users, target accounts, terminal sessions, commands executed with elevated privileges, and sudo session activity.

The dataset represents a laboratory snapshot rather than a continuous production log stream. Therefore, conclusions drawn from the investigation are limited to the activity contained within the available dataset.

2.4 Authorization and Scope

All activities documented in this report were performed within a controlled local laboratory environment for cybersecurity learning, security monitoring, and incident investigation purposes.

No unauthorized systems or third-party environments were targeted during the assessment.

3. TOOLS USED

The following tools and technologies were used during the assessment:

Tool / Technology	Purpose
Splunk Enterprise 10.4.3	SIEM platform used for log ingestion, search, monitoring, detection, and investigation.
Kali Linux	Security-focused operating system used as the laboratory environment.
VirtualBox	Virtualization platform used to host the Kali Linux laboratory.
Splunk Search Processing Language (SPL)	Used to query, filter, extract, aggregate, and analyse security events.
Linux journalctl	Used to collect sudo and PAM-related system activity from the Linux system journal.
Linux sudo / PAM logs	Source data for authentication and privilege-activity investigation.
Nessus	Security assessment tool whose administrative activity appeared within the laboratory log dataset.
GitHub	Used to store and present the completed project documentation and supporting evidence

3.1 PRIMARY TECHNOLOGIES

The assessment primarily focused on Splunk Enterprise, SPL, Linux authentication and privilege events, and SOC investigation techniques.

Splunk served as the central SIEM platform, while the Linux journal provided the source security events used for analysis. SPL queries were used throughout the investigation to identify patterns, extract relevant fields, summarize activity, and establish timelines.

4. PROJECT TYPE

This project is classified as a: **SIEM / SOC Security Monitoring & Incident Investigation**

The project focuses on using a Security Information and Event Management (SIEM) platform to collect, search, monitor, analyse, and investigate security-related events within a controlled laboratory environment.

The investigation emphasizes privilege activity monitoring, authentication-event analysis, timeline reconstruction, detection creation, alert monitoring, and evidence-based security assessment.

5. OBJECTIVES

The primary objectives of this project were to:

1. Deploy and configure Splunk Enterprise within a Kali Linux laboratory environment.
2. Collect and prepare Linux authentication and privileged-activity logs for SIEM analysis.
3. Ingest security events into a dedicated Splunk index.
4. Develop and execute SPL queries to identify meaningful security events and activity patterns.
5. Identify and analyse root-level privileged activity.
6. Examine command execution, target accounts, initiating users, and sudo session activity.
7. Establish timelines for privileged activity and investigate periods of increased activity.
8. Create a Splunk detection alert for root-level privilege activity.
9. Examine alert trigger activity and investigate detected events.
10. Assess whether observed activity was consistent with authorized laboratory administration or indicated potentially malicious behavior.
11. Document findings, security observations, and appropriate remediation recommendations.
12. Demonstrate practical junior SOC analyst skills in SIEM monitoring, log analysis, detection, investigation, and security reporting.

6. METHODOLOGY

The assessment followed a structured SIEM monitoring and incident investigation workflow. The methodology was designed to simulate the activities a junior SOC analyst would perform when collecting security data, identifying potentially significant events, investigating activity, and documenting findings.

6.1 Environment Preparation

A Kali Linux virtual machine was configured within VirtualBox as the primary security monitoring laboratory environment. Splunk Enterprise 10.4.3 was installed and configured on the virtual machine.

The laboratory environment was allocated additional memory to support Splunk operations and maintain stable performance during log analysis.

6.2 Security Log Collection

Linux system activity was collected from the system journal using "journalctl". Sudo and PAM-related events were filtered to create a focused dataset containing authentication and privileged-command activity.

The resulting dataset was stored as:

```
"/var/log/splunk-lab/splunk_auth.log"
```

6.3 SIEM Log Ingestion and Corrupted Bucket Recovery

The prepared security-log file was initially configured as a monitored input in Splunk and indexed in the default "main" index. The initial ingestion successfully produced 215 events.

During subsequent analysis, Splunk began reporting bucket-read errors when querying the "main" index. The errors indicated that the affected bucket contained inconsistent or missing indexed data, including a reported mismatch between the "tsidx" event count and the available slice data.

Splunk's filesystem consistency check ("fsck") was used to safely investigate the affected bucket. The scan reported:

- Event-count mismatch between the index metadata and event data
- Missing Bloom filter data
- Raw event data unavailable for the affected bucket

A controlled repair attempt was performed using Splunk's bucket-repair functionality. The repair operation could not successfully reconstruct the affected bucket because the underlying event data required for recovery was unavailable.

The original security-log source file remained intact throughout the recovery process. Its contents were independently verified to contain 215 log lines, confirming that the source dataset had not been lost.

Rather than deleting the affected bucket or continuing with potentially destructive recovery actions, a new dedicated Splunk index named "splunk_lab" was created. The existing monitor was removed from the corrupted "main" index configuration and re-added with "splunk_lab" as its destination.

The source log was then successfully re-indexed into the new index, and Splunk returned 215 events from the recovered laboratory dataset.

This recovery approach preserved the original source data while avoiding unnecessary modification or deletion of the corrupted bucket. The subsequent investigation was therefore conducted using the verified "splunk_lab" dataset.

6.4 Data Analysis and Field Extraction

SPL queries were developed to extract and analyse relevant fields from the raw security events.

The investigation focused on:

- Initiating users
- Target accounts
- Executed commands
- Authentication sessions
- Privileged activity
- Event timestamps
- Command categories
- Service-management activity
- Package-management activity
- System-management activity

Regular-expression field extraction was used where necessary to transform information contained within raw log messages into searchable fields.

6.5 Privileged Activity Investigation

Root-targeted activity was isolated to identify commands executed with elevated privileges. Statistical searches were then used to determine activity volume, command frequency, target accounts, and command categories.

The investigation identified 67 root-targeted events involving 36 distinct commands.

6.6 Timeline Analysis

Time-based SPL searches were used to establish the investigation window and identify periods of increased privileged activity.

Hourly and daily timelines were examined, followed by focused investigation of the busiest period. The earliest and most recent root-level activities were also reviewed to establish the overall period represented by the dataset.

6.7 Detection and Alert Creation

A Splunk detection alert named “Root Privilege Activity Detection” was created to identify root-level privilege activity.

The alert was configured as a real-time detection with a trigger condition of more than zero matching results within one minute. Trigger history was reviewed to verify that the detection was being activated by matching activity.

6.8 Investigation and Assessment

Detected activity was reviewed in context rather than automatically classified as malicious.

Commands, timestamps, target accounts, and surrounding laboratory activity were considered when assessing whether observed behaviour was consistent with authorized system administration, security-tool configuration, maintenance, or potentially suspicious activity.

6.9 Evidence Collection and Documentation

Relevant Splunk search results and configuration screens were captured as supporting evidence throughout the investigation.

The evidence was organized separately from the written report and numbered for reference. The final project documentation links the analytical findings to the corresponding evidence items stored within the project's GitHub repository.

6.10 Findings and Recommendations

The results of the investigation were reviewed to identify security observations and determine appropriate recommendations.

Conclusions were based only on the available laboratory dataset. Where the available evidence was insufficient to establish malicious activity, the activity was documented as observed or investigated rather than being presented as a confirmed security incident.

7. SPLUNK INSTALLATION & CONFIGURATION

7.1 Splunk Installation

Splunk Enterprise was installed on the Kali Linux virtual machine used for the laboratory environment.

The Splunk Enterprise package installed during the assessment was:

- Version: 10.4.3
- Build: 4174a2deda5d
- Installation Directory: "/opt/splunk"
- Operating User: "splunk"

The installation package was transferred into the Kali Linux environment and installed using the Debian package manager.

The installation was verified after completion by checking the installed Splunk package and confirming the reported Splunk version.

7.2 Splunk Service Configuration

Splunk was configured to operate under the dedicated "splunk" service account rather than running the Splunk service directly as root.

The Splunk service was started and stopped using the Splunk command-line interface. The Splunk Web interface was made available on port 8000 for security monitoring and investigation.

7.3 Resource Configuration

During the initial setup, the Kali Linux virtual machine experienced performance limitations while Splunk was running.

The virtual machine was subsequently allocated 4 GB of RAM to provide additional resources for Splunk operations and improve laboratory performance.

After the adjustment, the system provided approximately 3.8 GB of usable memory to the virtual machine, with no active swap usage observed during the verification.

7.4 Splunk Web Interface

The Splunk Web interface was accessed locally through:

"http://127.0.0.1:8000"

The Splunk dashboard was successfully accessed using the configured administrative account.

7.5 Installation Verification

The completed installation was verified using the Splunk command-line interface.

The verification confirmed:

- Splunk Enterprise was installed successfully.
- Splunk version 10.4.3 was available.
- The Splunk installation directory existed at `"/opt/splunk"`.
- Splunk services could be controlled using the dedicated `"splunk"` account.
- The Splunk Web interface was accessible on port 8000.

The installation and configuration established the SIEM platform required for subsequent log ingestion, security monitoring, SPL-based investigation, and alert configuration.

8. SIEM CONFIGURATION & LOG INGESTION

8.1 Security Log Preparation

The Linux system journal was used as the primary source of security events for the investigation.

Sudo-related events were extracted from the journal using `"journalctl"` and filtered to create a focused authentication and privileged-activity dataset.

The resulting log file was stored in:

`"/var/log/splunk-lab/splunk_auth.log"`

The dedicated directory was used to provide Splunk with appropriate access to the log source while keeping the dataset separate from other system log locations.

8.2 Splunk Data Input Configuration

The prepared log file was configured as a monitored input in Splunk.

The monitored source was:

`"/var/log/splunk-lab/splunk_auth.log"`

The input was ultimately configured to send events to the dedicated laboratory index:

`"index=splunk_lab"`

This provided a separate location for the investigation dataset and prevented the subsequent analysis from depending on the previously affected `"main"` index bucket.

8.3 Data Verification

After configuring the monitored input, Splunk Search & Reporting was used to verify that the security-log dataset had been successfully indexed.

The following SPL query was used:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
```

The query returned 215 events, confirming that the source file was successfully ingested into the "splunk_lab" index.

The search results also confirmed the expected source:

```
"/var/log/splunk-lab/splunk_auth.log"
```

and provided the event data used for the subsequent security investigations.

This verification established that the new laboratory index was receiving and returning the complete 215-event dataset before further SPL analysis was performed.

8.4 Event Structure

The ingested events retained useful information from the original Linux journal entries, including:

- Event timestamp
- Initiating user
- Terminal session
- Working directory
- Target user
- Command executed
- Sudo session activity

This information provided the foundation for subsequent SPL-based field extraction and investigation.

8.5 Search and Reporting Validation

Splunk Search & Reporting was used to verify that events could be retrieved from the new laboratory index.

The successful retrieval of the 215-event dataset confirmed that the monitored input, index configuration, and source log were functioning correctly.

The verified dataset was then used for privileged-activity analysis, timeline reconstruction, user activity analysis, and detection engineering.

9. SECURITY MONITORING & SPL QUERIES

Splunk Search Processing Language (SPL) was used throughout the investigation to transform the raw security events into meaningful security information.

The queries were designed to answer common SOC investigation questions, including:

- How many security events were collected?
- Which accounts were targeted by privileged commands?
- Which commands were executed with root privileges?
- Which command categories generated the most activity?
- When did privileged activity occur?
- Which user initiated the activity?
- Were there unusual system-management or account-modification events?
- Can a detection alert identify root-level activity?

The following subsections document the principal SPL searches used during the investigation.

9.1 Baseline Event Verification

The initial verification query confirmed that the laboratory dataset was available in Splunk:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
```

The search returned 215 events.

This established the baseline dataset used throughout the investigation.

9.2 Extracting Privileged Activity Fields

The raw sudo events contained information about the target account and command executed. SPL field extraction was used to make these values searchable.

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
```

```
| rex "USER=(?<target_user>[^;]+)"
```

```
| rex "COMMAND=(?<command>.*)"
```

The extracted fields "target_user" and "command" were subsequently used to filter and analyze privileged activity.

9.3 Identifying Root Privilege Activity

To identify events where commands were executed against the root account, the following query was used:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root"
| stats count by target_user
```

The search identified 67 root-targeted events.

This provided the primary dataset for the privileged-activity investigation.

9.4 Privileged Activity by Target Account

The following query was used to compare privileged activity across target accounts:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^;]+)"
| stats count by target_user
| sort - count
```

The results identified:

- root: 67 events
- splunk: 5 events

This showed that root-targeted activity represented the majority of the extracted privileged-command activity in the dataset.

9.5 Privileged Activity by Command

The following query was used to identify individual commands associated with root-level activity:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root"
| stats count by command
| sort - count
```

This search returned 67 root-targeted events across 36 distinct commands.

The results provided a detailed view of the commands executed with elevated privileges and formed the basis for subsequent command-category analysis.

9.6 Command Category Analysis

To group root-level commands by their executable or command category, the following SPL query was used:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
```

```
| rex "USER=(?<target_user>[^;]+)"
```

```
| rex "COMMAND=(?<command>.*)"
```

```
| search target_user="root"
```

```
| rex field=command "^/[^ ]*/(?<program>[^/]+)"
```

```
| stats count by program
```

```
| sort - count
```

The resulting command categories included system service management, package management, Nessus administration, system-log inspection, reboot operations, file inspection, and account-management activity.

The observed counts were:

- "systemctl" — 20
- "apt" — 14
- "nessuscli" — 10
- "journalctl" — 5
- "reboot" — 5
- "tail" — 5
- "dpkg" — 4
- "nano" — 1
- "su" — 1
- "usermod" — 1

These results provided a high-level view of how elevated privileges were being used within the laboratory environment.

10. PRIVILEGE ACTIVITY INVESTIGATION

Privileged activity was selected as the primary investigation focus because commands executed with elevated privileges can significantly affect system configuration, services, users, packages, and security controls.

The investigation used SPL to identify root-targeted activity, determine the volume and nature of the activity, and establish when the activity occurred.

10.1 Root Privilege Activity

The investigation identified 67 events in which the target account was "root".

The following SPL query was used:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
```

```
| rex "USER=(?<target_user>[^\;]+)"
```

```
| rex "COMMAND=(?<command>.*)"
```

```
| search target_user="root"
```

```
| stats count by target_user
```

The result established 67 root-targeted privilege events within the available dataset.

Root-level activity was not automatically treated as malicious. Each activity type was examined in context to determine whether it appeared consistent with authorized laboratory administration or required further investigation.

10.2 Root Activity by Command Category

The command-category analysis identified the following distribution:

Command Category	Events
systemctl	20
apt	14
nessuscli	10
journald	5
reboot	5
tail	5
dpkg	4
nano	1
su	1
usermod	1

The results showed that the majority of root-level activity involved system administration, package management, service management, security-tool administration, and system-log inspection.

The presence of elevated commands alone was therefore insufficient to establish malicious behaviour.

10.3 System Service Management

A focused investigation was performed against "systemctl" activity to identify privileged service-management operations.

The following SPL query was used:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root" command="*/systemctl *"
| table _time target_user command
| sort _time
```

The results included service status checks and operations involving services such as Nessus and VirtualBox Guest Utilities.

Service-management activity can be security-relevant because privileged users can start, stop, restart, or modify services. In this laboratory dataset, the observed operations were consistent with the configuration and troubleshooting of security and virtualization software.

10.4 Package Management Activity

Package-management commands executed with root privileges were also investigated.

The following SPL query was used:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root" (command="*/apt *" OR command="*/dpkg *")
| table _time target_user command
| sort _time
```

The results included package updates, package installation, package removal, and package-status operations.

Package-management activity can be important during a security investigation because attackers may install tools, remove security software, or modify system components. However, the commands observed in this dataset were associated with the laboratory's system and VirtualBox configuration activities.

10.5 System Reboot Activity

Root-level reboot operations were investigated using:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^\;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root" command="*/reboot*"
| table _time target_user command
| sort _time
```

Five root-level reboot events were identified within the dataset.

The events occurred during the laboratory's system-management period. Reboot activity was therefore documented as an observed privileged action rather than classified as malicious activity.

10.6 Account and Group Modification

Account-management activity was also examined because changes to user privileges or group membership can have security implications.

The following SPL query was used:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^\;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root"
| search command="*user*" OR command="*passwd*" OR command="*group*"
| table _time target_user command
| sort _time
```

The investigation identified a root-level "usermod" command that added the Kali user to the "vboxsf" group.

This activity was associated with VirtualBox shared-folder configuration within the laboratory environment. Although group-membership changes are security-relevant and should be monitored in production environments, the available context did not establish this event as malicious.

10.7 System Log Inspection

Root-level "journalctl" activity was also examined:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
```

```
| rex "USER=(?<target_user>[^;]+)"
```

```
| rex "COMMAND=(?<command>.*)"
```

```
| search target_user="root" command="*/journalctl*"
```

```
| table _time target_user command
```

```
| sort _time
```

Five root-level system-log inspection events were identified.

Log inspection is a normal administrative and troubleshooting activity but is also relevant to SOC investigations because unauthorized users may attempt to review system logs while investigating or attempting to conceal activity.

Within this dataset, the observed log-inspection commands were consistent with the laboratory troubleshooting and investigation workflow.

10.8 Investigation Assessment

The privileged-activity investigation identified a substantial amount of root-level activity, but the nature and context of the commands were important when assessing their significance.

The available evidence showed activity primarily associated with:

- System administration
- Splunk configuration
- Nessus administration
- Package management
- VirtualBox configuration
- Service management
- System-log inspection

No single event or command provided sufficient evidence to establish a confirmed compromise.

The investigation therefore classified the observed root-level activity as security-relevant privileged activity requiring monitoring, while avoiding an unsupported conclusion that the activity represented a confirmed security incident.

11. TIMELINE ANALYSIS

A timeline-based investigation was performed to understand the distribution of privileged activity across the period represented by the laboratory dataset.

Time-based SPL searches were used to identify the earliest and most recent root-level activity, examine daily and hourly activity patterns, and identify the period with the highest concentration of privileged events.

11.1 Overall Investigation Window

The following SPL query was used to establish the earliest and latest root-level activity:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root"
| stats earliest(_time) as first_activity latest(_time) as last_activity count as total_events
| convert ctime(first_activity) ctime(last_activity)
```

The results identified:

- Total root-targeted events: 67
- First recorded root activity: 09/02/2026 13:10:16
- Last recorded root activity: 09/14/2026 07:26:19

This established the overall investigation window represented by the available root-privilege events.

11.2 Daily Privileged Activity

Daily activity was analysed using the following query:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root"
| timechart span=1d count
```

The resulting timeline showed how root-level activity was distributed across individual days.

Daily analysis is useful for identifying changes in activity volume and determining whether privileged operations were concentrated within particular periods.

11.3 Hourly Privileged Activity

A more granular hourly timeline was generated using:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root"
| timechart span=1h count
```

The hourly view provided greater visibility into short periods of increased privileged activity.

11.4 Identification of the Busiest Activity Period

The following SPL query was used to identify the hours containing the highest number of root-targeted events:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root"
| bin _time span=1h
| stats count by _time
| sort - count
| head 10
```

The investigation identified:

08 September 2026 at 15:00 — 16 root-targeted events

This represented the highest concentration of root-level activity within the available dataset.

11.5 Investigation of the Busiest Hour

The busiest hour was investigated using a time-bounded query:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^;]+)"
| rex "COMMAND=(?<command>.*)"
```

```
| search target_user="root"
| where _time >= strptime("2026-09-08 15:00:00", "%Y-%m-%d %H:%M:%S")
AND _time < strptime("2026-09-08 16:00:00", "%Y-%m-%d %H:%M:%S")
| stats count by command
| sort - count
```

The search returned 16 events across 12 command statistics.

The activity during this period included Nessus registration and update operations, service-management commands, security-tool status checks, log inspection, and system configuration activity.

Because the commands were examined in the context of the authorized laboratory environment, the concentrated activity was not treated as evidence of a confirmed attack.

11.6 Earliest Root Activity

The first ten root-targeted events were examined using:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
| rex "USER=(?<target_user>[^\;]+)"
| rex "COMMAND=(?<command>.*)"
| search target_user="root"
| table _time target_user command
| sort _time
| head 10
```

The earliest activity included:

- "su"
- "VBoxLinuxAdditions.run"
- "reboot"
- "apt update"
- Package installation
- VirtualBox Guest Utilities service management
- Package removal and updates

This sequence was consistent with the laboratory's initial system and VirtualBox configuration activities.

11.7 Most Recent Root Activity

The most recent root-targeted events were examined using:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
```

```
| rex "USER=(?<target_user>[^;]+)"
```

```
| rex "COMMAND=(?<command>.*)"
```

```
| search target_user="root"
```

```
| table _time target_user command
```

```
| sort - _time
```

```
| head 10
```

The most recent activity included repeated "journalctl" commands, Splunk package verification and installation activity, a reboot, VirtualBox group configuration, and other system-management operations.

The most recent recorded root event was a "journalctl" command at 09/14/2026 07:26:19.

11.8 Timeline Assessment

The timeline investigation demonstrated that privileged activity occurred throughout the period represented by the dataset, with a notable concentration during the identified busiest hour.

The temporal concentration was investigated rather than automatically classified as suspicious. Examination of the associated commands showed activity largely related to system administration, security-tool configuration, and laboratory troubleshooting.

The timeline therefore provided useful context for the investigation but, by itself, did not establish malicious behaviour.

12. ALERT CREATION & DETECTION

A Splunk detection alert was created to identify root-level privileged activity within the laboratory environment.

The purpose of the alert was to demonstrate how a SOC analyst can convert an identified security-relevant behaviour into an automated detection.

12.1 Detection Objective

The detection was designed to identify events where a command was executed with the target account set to "root".

The detection logic was based on the same field extraction and filtering approach used during the privileged-activity investigation.

The core detection query was:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
```

```
| rex "USER=(?<target_user>[^\;]+)"
```

```
| rex "COMMAND=(?<command>.*)"
```

```
| search target_user="root"
```

This query identifies root-targeted privileged events and provides the extracted command for further investigation.

12.2 Alert Configuration

The Splunk alert was configured with the following settings:

Configuration	Value
Alert Name	Root Privilege Activity Detection
Description	Detects root-level privilege command activity in the Splunk lab environment for SOC monitoring and investigation
Alert Type	Real-time
Trigger Condition	Number of Results >0
Trigger Window	1 minute
Status	Enabled
Action	Add to Triggered Alerts
Permissions	Private

The alert was configured to trigger whenever at least one matching root-level privilege event was detected within the defined one-minute window.

12.3 Detection Validation

After the alert was enabled, its trigger history was reviewed.

Multiple trigger-history entries were observed, demonstrating that the alert was being activated when matching events were encountered.

The underlying detection search was also reviewed using the full available dataset. The search returned the previously identified root-level activity, confirming that the detection logic was matching the expected events.

12.4 Detection Significance

The alert demonstrates a basic detection-engineering workflow:

1. Identify a security-relevant behaviour.
2. Develop an SPL query to detect the behaviour.
3. Convert the query into an automated alert.
4. Define an appropriate trigger condition.
5. Enable the detection.
6. Monitor trigger history.
7. Investigate the activity that caused the detection.

In a production SOC environment, root-level privilege activity would generally require contextual investigation because legitimate administrators and security tools may also perform privileged operations.

The alert therefore serves as a detection mechanism, rather than automatically determining that every matching event represents malicious activity.

13. ALERT INVESTIGATION

Following the creation and activation of the Root Privilege Activity Detection alert, the alert trigger history was reviewed to investigate the activity responsible for generating detections.

13.1 Alert Trigger History

The Splunk alert history displayed multiple trigger events after the detection was enabled.

The trigger history provided timestamps for individual alert activations and included options to review the associated search results.

This demonstrated that the configured detection was actively evaluating matching events and generating alert activity.

13.2 Investigation of Triggered Activity

The underlying SPL detection query was reviewed against the available laboratory dataset to identify the privileged activity responsible for the detections.

The detection query was:

```
index=splunk_lab source="/var/log/splunk-lab/splunk_auth.log"
```

```
| rex "USER=(?<target_user>[^;]+)"
```

```
| rex "COMMAND=(?<command>.*)"
```

```
| search target_user="root"
```

The search identified the previously established 67 root-targeted events.

The events were then examined according to their timestamps, commands, target accounts, and surrounding laboratory activity.

13.3 Alert Result Time-Range Consideration

During the investigation, selecting View Results from individual alert-history entries initially produced no matching results when a narrow time range was applied.

The search was subsequently reviewed using the full available time range, where the underlying detection query returned the expected root-level activity.

This behaviour was documented as a limitation of investigating alerts against the available static laboratory dataset. The alert trigger history demonstrated that the alert had fired, while the selected search time range affected whether the associated historical events were visible during manual review.

13.4 Investigation Outcome

The alert successfully identified root-level privileged activity, which was then correlated with the broader investigation.

The activity included:

- Security-tool administration
- System service management
- Package management
- System-log inspection
- System configuration
- VirtualBox-related administration
- Splunk installation and configuration

The available evidence did not establish a confirmed malicious compromise.

The alert was therefore treated as a security-relevant detection requiring analyst review, rather than as automatic confirmation of an incident.

13.5 SOC Investigation Approach

The investigation followed a basic SOC workflow:

Alert → Validate → Review Event → Extract Context → Correlate Activity → Assess Risk → Document Finding

This approach helps reduce false conclusions by requiring contextual analysis before classifying privileged activity as malicious.

14. FINDINGS & RISK ASSESSMENT

The investigation identified several security-relevant observations within the laboratory dataset. These findings were assessed based on their observed activity, potential security significance, and the available context.

The findings below represent observations from the laboratory dataset and should not be interpreted as evidence of confirmed malicious activity unless explicitly stated.

14.1 Finding 1 — Root-Level Privileged Activity

Severity: Medium

Observation: The investigation identified 67 root-targeted privilege events involving 36 distinct commands.

Security Significance: Root-level access provides extensive control over a Linux system. Unexpected or unauthorized use of root privileges can allow an attacker to modify system configuration, install software, manipulate services, access sensitive files, or interfere with security controls.

Assessment: The observed root activity was predominantly associated with laboratory administration, system maintenance, Splunk configuration, Nessus administration, package management, and VirtualBox configuration.

No confirmed malicious root activity was established from the available dataset.

Recommendation: Organizations should monitor privileged-account activity and investigate unexpected root-level commands, particularly when they occur outside approved maintenance periods or are associated with unfamiliar processes or users.

14.2 Finding 2 — Concentrated Privileged Activity

Severity: Medium

Observation: The busiest identified period occurred on 08 September 2026 at 15:00, with 16 root-targeted events recorded within the hour.

Security Significance: A sudden concentration of privileged operations can warrant investigation because attackers may perform multiple administrative actions during a compromise.

Assessment: The 16-event period was investigated at the command level. The activity included Nessus administration, service management, log inspection, and system configuration.

The available context was consistent with authorized cybersecurity-tool administration and laboratory troubleshooting.

Recommendation: SOC teams should investigate bursts of privileged activity by correlating timestamps with maintenance schedules, change records, user identity, source systems, and other security events.

14.3 Finding 3 — Root-Level Service Management

Severity: Medium

Observation: The dataset contained privileged "systemctl" operations, including service status checks, service starts, stops, and restarts.

Security Significance: Privileged service-management commands can affect the availability and security of critical services. Unauthorized service manipulation can potentially disable security controls or establish persistence.

Assessment: The observed service-management activity included Nessus and VirtualBox Guest Utilities operations and was consistent with the laboratory configuration and troubleshooting activities.

Recommendation: Monitor privileged service-management commands and investigate unexpected changes to security-related services.

14.4 Finding 4 — Root-Level Package Management

Severity: Medium

Observation: The investigation identified privileged "apt" and "dpkg" activity involving package updates, installation, removal, and package-status checks.

Security Significance: Package-management activity can introduce software changes to a system. Unauthorized installation or removal of packages can create security risks or weaken existing controls.

Assessment: The observed package-management activity was associated with Linux and VirtualBox configuration within the laboratory environment.

Recommendation: Organizations should monitor package installation and removal activity and correlate significant changes with approved administrative or change-management records.

14.5 Finding 5 — Root-Level Account Group Modification

Severity: Medium

Observation: One root-level "usermod" command modified the Kali user's group membership by adding the user to the "vboxsf" group.

Security Significance: Changes to user group membership can alter access privileges and should therefore be monitored.

Assessment: The observed modification was associated with VirtualBox shared-folder configuration in the laboratory environment. No evidence from the available dataset established that the modification was unauthorized.

Recommendation: Monitor changes to privileged or security-sensitive groups and validate them against approved administrative activity.

14.6 Finding 6 — Root-Level System Log Inspection

Severity: Low

Observation: Five root-level "journalctl" events were identified.

Security Significance: System-log inspection is a normal administrative activity, but unexpected access to logs can also be relevant during an investigation.

Assessment: The observed activity was consistent with system troubleshooting and security investigation within the laboratory.

Recommendation: Monitor privileged access to security logs and correlate log-inspection activity with the identity, purpose, and timing of the administrator's actions.

14.7 Overall Risk Assessment

The investigation identified multiple security-relevant privileged activities, but the available evidence did not establish a confirmed compromise or malicious intrusion.

The majority of observed activity was consistent with authorized laboratory administration, cybersecurity-tool configuration, system maintenance, and troubleshooting.

The primary security lesson from the investigation is that privileged activity should be detected and investigated based on context rather than automatically classified as malicious.

The Splunk detection created during the project provides a foundation for monitoring this type of activity and could be expanded in a production environment through additional context such as user identity, source host, authentication method, command allowlists, maintenance windows, and correlation with other security events.

15. REMEDIATION RECOMMENDATIONS

Based on the observations identified during the investigation, the following recommendations are proposed to strengthen privileged-activity monitoring and improve the organization's ability to detect and investigate potentially unauthorized activity.

15.1 Monitor Privileged Account Activity

Organizations should continuously monitor activity involving privileged accounts such as "root" and other administrative accounts.

Detection rules should identify privileged commands and generate alerts when activity deviates from expected administrative behaviour.

15.2 Apply Least-Privilege Principles

Administrative privileges should be granted only when required for an approved task.

Where possible, users and service accounts should operate with the minimum permissions necessary to perform their functions.

15.3 Monitor Account and Group Changes

Changes to user accounts, group memberships, and privilege assignments should be monitored and correlated with authorized administrative activity.

Particular attention should be given to changes involving privileged or security-sensitive groups.

15.4 Monitor Security-Service Changes

SOC monitoring should include privileged actions that start, stop, restart, or modify security-related services.

Unexpected changes to security tooling or monitoring services should be investigated promptly.

15.5 Monitor Package Installation and Removal

Package-management activity should be monitored to identify unexpected installation, removal, or modification of software.

Where applicable, package changes should be correlated with approved maintenance activities or change-management records.

15.6 Improve Alert Context

The root-privilege detection developed during this project could be enhanced by incorporating additional contextual information, including:

- Initiating username
- Source host
- Authentication method
- Command executed
- Parent process
- Time of activity
- Known maintenance windows
- User role
- Historical activity patterns

Additional context can help reduce false positives and allow analysts to distinguish legitimate administrative operations from potentially unauthorized activity.

15.7 Maintain Centralized Security Logging

Security-relevant authentication and privileged-activity logs should be centralized in a SIEM platform and retained according to organizational requirements.

Centralized logging improves visibility, investigation capability, and timeline reconstruction during security incidents.

15.8 Protect SIEM Data Integrity

SIEM administrators should monitor the health of indexed data and investigate bucket or index corruption promptly.

Regular backups of important source logs should be maintained so that an indexing failure does not result in permanent loss of investigative data.

During this project, preservation of the original log source allowed the 215-event dataset to be re-indexed after corruption affected the initial Splunk index.

15.9 Develop Additional Detection Rules

The detection created in this project can serve as a foundation for additional SOC detections, including:

- Unexpected privileged-account usage
- Repeated failed authentication attempts
- Privileged account creation
- Unauthorized group membership changes

- Security-service stoppage
- Suspicious package installation
- Unusual command execution
- Privileged activity outside expected maintenance periods

These detections could be correlated within Splunk to provide a more comprehensive security-monitoring capability.

16. EVIDENCE SUMMARY

The investigation was supported by 31 evidence screenshots captured during the Splunk deployment, log ingestion, security monitoring, privileged activity investigation, timeline analysis, and alert configuration processes.

The evidence includes screenshots demonstrating Splunk installation and configuration, successful ingestion of 215 security events, privileged activity involving the root account, command and program analysis, timeline analysis, package and service activity, account-group modification, system-log investigation, and the configuration and triggering of the root privilege detection alert.

The evidence files are maintained separately from this report in the project's GitHub repository under:

Evidence/Screenshots/

Each screenshot is assigned an evidence number corresponding to the relevant investigation stage and is referenced throughout the report where applicable.

Evidence was retained separately from the PDF to maintain a clean report structure while preserving supporting technical documentation.

17. LIMITATIONS

This assessment was conducted within an authorized local cybersecurity laboratory using Kali Linux, VirtualBox, and a controlled Splunk environment. The analysis was based primarily on a static snapshot of Linux authentication and privileged-activity logs rather than a continuously monitored production environment.

The dataset was limited to 215 events, and therefore the findings may not represent the volume, complexity, or diversity of activity found in an enterprise environment.

The assessment also focused on selected privileged and authentication-related activities. Additional data sources such as endpoint telemetry, network traffic, Windows event logs, cloud logs, and threat-intelligence feeds were not incorporated into this investigation.

The results should therefore be interpreted as a practical demonstration of SIEM monitoring and SOC investigation techniques, rather than a complete assessment of an enterprise security environment.

18. CONCLUSION

This project demonstrated a practical approach to using Splunk Enterprise as a SIEM platform for security monitoring and incident investigation within an authorized local cybersecurity laboratory.

The assessment covered log collection and ingestion, SPL-based security monitoring, privileged activity analysis, timeline investigation, alert creation, alert validation, and evidence-based risk assessment. A total of 215 events were ingested into the dedicated `splunk_lab` index, including 67 root-targeted privileged activity events across 36 distinct commands.

The investigation identified several security-relevant activities, including privileged service management, package management, system-log inspection, reboot activity, and an account-group modification. After reviewing the available context, no confirmed malicious activity or compromise was identified within the analysed dataset.

The project also demonstrated the importance of preserving source logs and maintaining SIEM data integrity after encountering an initial Splunk bucket corruption issue. Rather than modifying the original source data destructively, the affected data was re-ingested into a fresh index and verified successfully.

Overall, the project strengthened practical junior SOC Analyst skills in SIEM administration, SPL querying, security monitoring, alert investigation, timeline analysis, evidence handling, and security documentation.

19. GITHUB PORTFOLIO REFERENCE

The complete project documentation and supporting evidence are maintained in the project's GitHub repository.

GitHub Repository:

<https://github.com/iwujidaniel/Cybersecurity-Portfolio>

Project:

Splunk-SIEM-Security-Monitoring-Incident-Investigation.

The repository contains the project report and supporting evidence screenshots demonstrating the Splunk configuration, log ingestion, security monitoring, privileged activity investigation, timeline analysis, and alert investigation performed during this assessment.

Evidence screenshots are stored separately in the project's *Evidence/Screenshots/* directory